



Disclaimer: This file is based completely on my opinions and experience. Everything you read here could be disagreed with. While I do recommend a team to know all of the systems written in this file, the order in which every system is written is merely a way to catalog them. I highly encourage teams to search about more systems that are not specified in this file.

Useful FRC systems ranked by system type:

Intakes

111 - Pivoting intake
112 - Cut bumper intake
120 - Under The Bumper
131 - Static Claw
132 - Passive claw
133 - Active Claw
140 - Rack and Pinion
151 - 3 Bar Linkage
152 - 4 Bar Linkage
153 - 6 Bar Linkage

Indexers

211 - Spinning Indexer
212 - Dual Spinning Indexer
221 - Belt Indexer
222 - Roller Indexer
230 - Dye Rotor Indexer
240 - Tunnel Indexer

Outtake

310 - Static Drum/Line Shooter
321 - Hooded Shooter
322 - Adjustable Line Shooter
331 - Passive Dumper
332 - Active Dumper



Extension Mechanisms

411 - Continuous Elevator

412 - Cascading Elevator <3L

413 - Cascading Elevator >3L

421 - Telescopic Arm

End Effector Manipulation

510 - Turret

521- 3DOF Arm

522 - 4+DOF Arm

Misc

910 - Ratchet

921 - Differential Drive

922 - Differential Gearbox

-  - Easy to understand and easy to model
-  - Medium level, harder concept and a more complicated model
-  - Hard concept to grasp, difficult to model
-  - Rarely practical, very difficult and usually not worth the hassle.